

POSITION DESCRIPTION AND ENGINEERING HIRING STANDARD (TAB C)

FICTIONAL SAMPLE. This document was created for software testing. All names, companies, schools, addresses, receipt numbers, and dates are invented.

CALDERON RIDGE ROBOTICS, INC. — Automation Engineering Group

Document CRR-HR-STD-014, revision 6, effective January 5, 2026

Owner: Vice President of Engineering | Approved by: President

PART 1 — POSITION DESCRIPTION: CONTROLS SYSTEMS ENGINEER II

| Field | Value |

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| Job title | Controls Systems Engineer II |

| Job code | ENG-CS-2 |

| Department | Automation Engineering |

| Reports to | Director of Automation Engineering |

| FLSA classification | Exempt |

| Salary band | \$98,000 – \$119,000 |

| Occupational classification | 17-2199, Engineers, All Other |

| Positions in band | 9 |

PURPOSE

The Controls Systems Engineer II owns the control architecture for assigned customer systems from concept through site acceptance. The role is accountable for the correctness of the motion design, the control software, the functional-safety design, and the commissioned result.

ESSENTIAL FUNCTIONS

1. Produce the control architecture document for each assigned system, including axis sizing calculations, network topology, cycle-time budget, and I/O allocation.
2. Develop control software in IEC 61131-3 Structured Text and function block diagram, under version control, with unit-tested function blocks.
3. Model and simulate the controlled mechanism; tune servo loops and document gain and phase margins before energising hardware.
4. Perform hazard and risk assessment; determine and verify the required performance level for each safety function to ISO 13849-1; produce and sign the safety validation report.
5. Integrate vision inspection and reject actuation, including camera-to-robot frame calibration.
6. Execute factory acceptance testing and site acceptance testing; resolve deviations; deliver as-built documentation.
7. Provide engineering direction to Automation Technicians and to Controls Systems Engineer I staff assigned to the project.

AUTHORITY RESERVED TO THIS LEVEL

- Approval of the control architecture document.
- Signature on the safety validation report.
- Approval to energise a system for the first time.

These three authorities are not delegated to the Engineer I level or to technicians.

MINIMUM EDUCATION REQUIREMENT

A bachelor's degree or higher in Mechatronics Engineering, Electrical Engineering, Control Systems Engineering, Robotics Engineering, or a closely related engineering specialty is required. Coursework must include control systems theory, dynamics, and either embedded real-time systems or industrial automation.

This requirement is mandatory. No combination of experience, vendor certification, or internal training substitutes for it. A candidate who does not hold a qualifying degree is not eligible for this job code and must be considered instead for job code TECH-AUT-3, Automation Technician III.

MINIMUM EXPERIENCE REQUIREMENT

Two years of controls engineering experience, of which at least one year involved multi-axis coordinated motion. Relevant graduate research or a graduate thesis in robotics or control systems may satisfy up to one year of this requirement. Experience never substitutes for the education requirement above.

PART 2 — COMPARISON OF LEVELS WITHIN THE OCCUPATIONAL CATEGORY

| | Automation Technician III (TECH-AUT-3) | Controls Systems Engineer I (ENG-CS-1) | Controls Systems Engineer II (ENG-CS-2) |

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| Minimum education | Associate degree in electrical or industrial technology | Bachelor's in a qualifying engineering specialty | Bachelor's or higher in a qualifying engineering specialty |

| Designs control architecture | No | No — implements an approved architecture | Yes — owns and approves it |

| Performs axis sizing and dynamic analysis | No | Assists | Yes |

| Performs ISO 13849-1 performance-level calculation | No | No | Yes |

| Signs safety validation report | No | No | Yes |

| Salary band | \$61,000 – \$74,000 | \$79,000 – \$96,000 | \$98,000 – \$119,000 |

PART 3 — DEGREE CREDENTIALS OF PERSONS WHO HAVE HELD JOB CODE ENG-CS-2

| Year hired | Status | Degree held at hire | Institution |

|---|---|---|---|

| 2018 | Current employee | B.S. Electrical Engineering | Kettleridge State University |

| 2019 | Current employee | M.S. Mechanical Engineering (control systems concentration) | Ambersley Polytechnic Institute |

| 2021 | Current employee | B.Eng. Mechatronics Engineering | Norwood Bay University |

| 2022 | Departed 2025 | B.S. Robotics Engineering | Fairmount Lake University |

| 2024 | Current employee | M.S. Control Systems Engineering | Kettleridge State University |

| 2025 | Current employee | B.S. Electrical Engineering | Halstrom University |

No person has ever been placed in job code ENG-CS-2 without a degree in one of the specialties listed in Part 1.

PART 4 — PARALLEL POSITIONS AT SIMILAR ORGANIZATIONS

The following announcements for parallel positions were collected from public postings between November 2025 and February 2026. Each is from a control-systems integrator of comparable size and scope of work.

Employer	Employees	Position advertised	Stated education requirement
Brantwick Motion Systems, LLC (Peoria, IL)	60	Controls Engineer II	"Bachelor's degree in Electrical, Mechatronics, or Controls Engineering required."
Odenfeld Integration Group, Inc. (Green Bay, WI)	95	Senior Automation Controls Engineer	"BS in Electrical Engineering, Mechatronics, or Robotics Engineering is required; equivalent experience will not be substituted."
Pell & Marrow Automation Co. (Springfield, MO)	44	Machine Controls Engineer	"Requires a four-year engineering degree in a controls-related discipline (EE, ME with controls emphasis, mechatronics)."
Suttergate Robotics Partners (Fort Wayne, IN)	130	Motion Control Engineer II	"Bachelor's or Master's in Robotics, Mechatronics, or Electrical Engineering required. Safety design to ISO 13849 required."
Vandermolt Systems, Inc. (Chattanooga, TN)	78	Controls & Safety Engineer	"Minimum: accredited BS in Mechatronics, Electrical, or Control Systems Engineering."

In addition, the *Association of Motion and Safety Integrators* 2025 Compensation and Practice Survey, at page 31, reports that 94 percent of responding member firms require a bachelor's degree in an engineering specialty for controls engineering roles above the entry level, and that 61 percent of holders of such roles hold a graduate degree. A copy of the relevant survey pages is attached to this document.

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